# CSE211s Introduction to Embedded Systems

Spring 2025

Binary System Review

1- Add the following binary and hexadecimal numbers:

| | | |
|---|---|---|
| 10010 | 1011101 | 10011 |
| + 1100 | +1000000 | +1111101 |
| | | |
| 0xA5 | 0xC4 | 0x46 |
| +0x83 | +0xDE | +0x62 |

2- How many times the binary number 10000 is greater than 10?

3- For the three systems [binary, decimal, hexa-decimal] convert each of the following numbers to the other two systems:

$00101110_2$

$720_{10}$

$FE0F_{16}$

4- Assume a 12-bit CPU: (Give answer in binary and hexadecimal)
What is the largest unsigned number?
What is the lowest unsigned number?
What is the largest signed number?
What is the lowest signed number?
Note: In a computer system that represents all integer quantities using two's complement form, the most significant bit has a negative place-weight. For example, an eight-bit system, the place weights are as follows:

$\underline{\phantom{xx}}\ \underline{\phantom{xx}}\ \underline{\phantom{xx}}\ \underline{\phantom{xx}}\ \underline{\phantom{xx}}\ \underline{\phantom{xx}}\ \underline{\phantom{xx}}\ \underline{\phantom{xx}}$

$-2^7 \quad 2^6 \quad 2^5 \quad 2^4 \quad 2^3 \quad 2^2 \quad 2^1 \quad 2^0$

5- Given the above place-weighting, convert the following eight-bit two's complement binary numbers into decimal form:

| | | |
|---|---|---|
| $01000101_2$ = | $01110000_2$ = | $11000001_2$ = |
| $10010111_2$ = | $01010101_2$ = | |

6- Add the following eight-bit two's complement numbers together, and then convert all binary quantities into decimal form to verify the accuracy of the addition:

| | | |
|---|---|---|
| 10110111 | 00111101 | 11111011 |
| +01110110 | +00111011 | +11111011 |
| | | |
| 10000001 | 01111011 | 01111111 |
| +10010001 | +00111101 | +10000001 |

7- How is it possible to tell that *overflow* has occurred in the addition of binary numbers, without converting the binary sums to decimal form and having a human being verify the answers?

8- Consider the addition of two signed 32-bit numbers. If a positive number is added to a negative number, under what conditions will a signed overflow occur?

9- Consider the subtraction of two signed 32-bit numbers. If a positive number is subtracted from another positive number, under what conditions will a signed overflow occur?

10- Let A and B be two 8-bit inputs to an 8-bit binary adder. Fill in the table showing R=A+B and the four flags after each addition. The first row illustrates the process.

| A | B | $R_{10}$ | $R_2$ | NZVC |
|---|---|---|---|---|
| 10 | 100 | 110 | | 0000 |
| 0x40 | 0xA2 | | | |
| 0xC3 | 0x6F | | | |
| 100 | -100 | | | |
| 110 | 146 | | | |
| 50 | | | | 0101 |

11- Let A and B be two 8-bit inputs to an 8-bit binary subtractor. Fill in the table showing R=A-B and the four flags after each subtraction. The first row illustrates the process. Calculate the carry bit in a similar way as the Cortex-M does.

| A | B | $R_{10}$ | $R_2$ | NZVC |
|---|---|---|---|---|
| 100 | 10 | 90 | | 0001 |
| 0x51 | 0x93 | | | |
| 0xDF | 0x9F | | | |
| -70 | -70 | | | |
| | 97 | | | 0101 |

12- Given the binary number A = 00001011, perform the bitwise logical operations:

$A \wedge$ 11110011

$A \vee$ 00001100

$A \oplus$ 00001100

Comment on the result.

13- Find the total amount of addressable memory, for each of the following CPUs, given the size of the address bus:

a) 16-bit address bus. (in K)
b) 24-bit address bus. (in megs)
c) 32-bit address bus. (in megabytes and gigabytes)
d) 48-bit address bus. (in megabytes, gigabytes, and terabytes)

14- What components are usually put together with the microcontroller onto a single chip?

15- In an embedded controller with on-chip ROM, why does the size of the ROM matter?

16- Given the memory map of the TM4C123:

a. What is the ending address for 256K flash ROM?
b. What is the starting address for 32K RAM?
c. What is the size of I/O ports registers block?